

Assignment No. 6

Ring.java

```
import java.util.Scanner;

class Process {
    int id;
    boolean active = true;
}

public class Ring {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of processes: ");
        int n = sc.nextInt();

        Process[] p = new Process[n];

        for (int i = 0; i < n; i++) {
            p[i] = new Process();
            System.out.print("Enter ID of process " + (i + 1) + ": ");
            p[i].id = sc.nextInt();
        }

        // simulate coordinator failure
        p[n - 1].active = false;
        System.out.println("Process " + p[n - 1].id + " is down");

        while (true) {
            System.out.println("\n1. Start Election\n2. Exit");
            int choice = sc.nextInt();
        }
    }
}
```

```

        if (choice == 2) break;

        System.out.print("Enter initiator process index (1 to " + n + "): ");
        int init = sc.nextInt() - 1;

        int current = init;
        int maxId = -1;

        System.out.println("\nElection started...");

        do {
            if (p[current].active) {
                System.out.println("Message passed to process " + p[current].id);

                if (p[current].id > maxId)
                    maxId = p[current].id;
            }

            current = (current + 1) % n;

        } while (current != init);

        System.out.println("\nNew Coordinator is Process " + maxId);
    }

    sc.close();
}
}

```

Output

C:\Users\anuja\OneDrive\Documents\DS_prac\Assignment5>notepad Ring.java

```
C:\Users\anuja\OneDrive\Documents\DS_prac\Assignment5>javac Ring.java
```

```
C:\Users\anuja\OneDrive\Documents\DS_prac\Assignment5>java Ring
```

```
Enter number of processes: 4
```

```
Enter ID of process 1: 1
```

```
Enter ID of process 2: 2
```

```
Enter ID of process 3: 3
```

```
Enter ID of process 4: 4
```

```
Process 4 is down
```

```
1. Start Election
```

```
2. Exit
```

```
1
```

```
Enter initiator process index (1 to 4): 3
```

```
Election started...
```

```
Message passed to process 3
```

```
Message passed to process 1
```

```
Message passed to process 2
```

```
New Coordinator is Process 3
```

```
1. Start Election
```

```
2. Exit
```

```
1
```

```
Enter initiator process index (1 to 4): 2
```

```
Election started...
```

```
Message passed to process 2
```

```
Message passed to process 3
```

```
Message passed to process 1
```

New Coordinator is Process 3

1. Start Election

2. Exit

2

Bully.java

```
import java.util.Scanner;
```

```
public class Bully {
```

```
    static boolean[] active;
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        int n = 5; // fixed 5 processes
```

```
        active = new boolean[n];
```

```
        // Initially all active
```

```
        for (int i = 0; i < n; i++) {
```

```
            active[i] = true;
```

```
        }
```

```
        System.out.println("Processes: P1 P2 P3 P4 P5");
```

```
        System.out.println("P5 is initial coordinator");
```

```
        while (true) {
```

```
            System.out.println("\n1. Down process\n2. Up process\n3. Send Election\n4. Exit");
```

```
            int choice = sc.nextInt();
```

```
            switch (choice) {
```

case 1:

```
System.out.print("Enter process to DOWN: ");  
int d = sc.nextInt();  
active[d - 1] = false;  
System.out.println("Process " + d + " is DOWN");  
break;
```

case 2:

```
System.out.print("Enter process to UP: ");  
int u = sc.nextInt();  
active[u - 1] = true;  
System.out.println("Process " + u + " is UP");  
break;
```

case 3:

```
System.out.print("Enter initiator process: ");  
int initiator = sc.nextInt();
```

```
if (!active[initiator - 1]) {  
    System.out.println("Process is DOWN");  
    break;  
}
```

```
boolean higherExists = false;
```

```
for (int i = initiator; i < n; i++) {  
    if (active[i]) {  
        System.out.println("Election message from P" + initiator + " to P" + (i + 1));  
        higherExists = true;  
    }  
}
```

```

    }

    if (!higherExists) {
        System.out.println("Process " + initiator + " becomes Coordinator");
    } else {
        for (int i = n - 1; i >= 0; i--) {
            if (active[i]) {
                System.out.println("Coordinator is P" + (i + 1));
                break;
            }
        }
    }
    break;

case 4:
    System.out.println("Program terminated...");
    return;
}
}
}
}

```

Output

C:\Users\anuja\OneDrive\Documents\DS_prac\Assignment6>notepad Bully.java

C:\Users\anuja\OneDrive\Documents\DS_prac\Assignment6>javac *.java

C:\Users\anuja\OneDrive\Documents\DS_prac\Assignment6>java Bully

Processes: P1 P2 P3 P4 P5

P5 is initial coordinator

1. Down process

2. Up process

3. Send Election

4. Exit

1

Enter process to DOWN: 5

Process 5 is DOWN

1. Down process

2. Up process

3. Send Election

4. Exit

3

Enter initiator process: 3

Election message from P3 to P4

Coordinator is P4

1. Down process

2. Up process

3. Send Election

4. Exit

4

Program terminated...